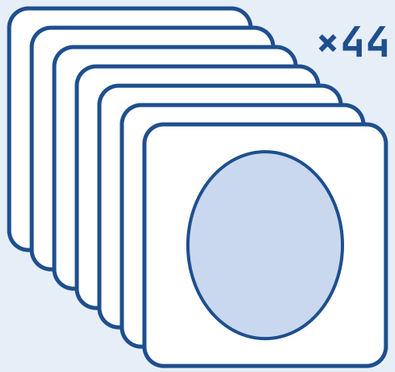
LESSON 3 · SCAN CONVERSION

DICOM to NIfTI in 4 Steps

Raw scanner files become one analysis-ready brain image, checked before anyone analyzes it

STEP 1

Raw Scanner Files



44 files · FLAIR_DICOM/

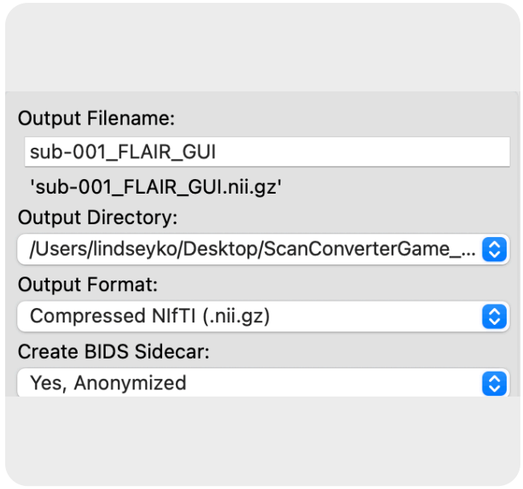
CONCEPT · DICOM SLICES

```
1.2.840.113619.6.495.64968897418720824957614882386
1.2.840.113619.6.495.64968897418720824957614882386
1.2.840.113619.6.495.64968897418720824957614882386
1.2.840.113619.6.495.64968897418720824957614882386
1.2.840.113619.6.495.64968897418720824957614882386
1.2.840.113619.6.495.64968897418720824957614882386
1.2.840.113619.6.495.64968897418720824957614882386
1.2.840.113619.6.495.64968897418720824957614882386
```

Scanners save one file per slice, named by machine.

STEP 2

Convert with dcm2niix



MRIcron · dcm2niix

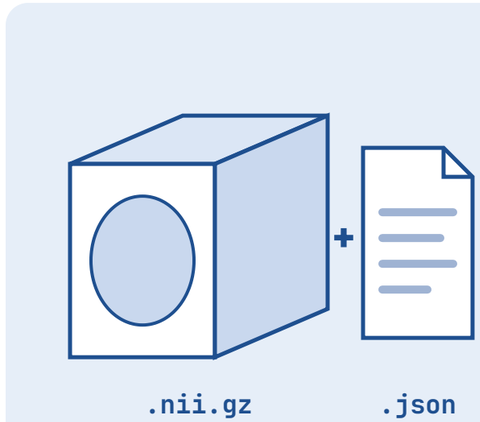
CONCEPT · FORMAT CONVERSION

```
/Applications/MRIcron.app/Contents/Resources/dcm2niix -f "sub-
Chris Rorden's dcm2niiX version v1.0.20201102  (JP2:OpenJPEG)
Found 44 DICOM file(s)
Convert 44 DICOM as /Users/lindseyko/Desktop/ScanConverterGa
Conversion required 44.028265 seconds (0.464288 for core cod
```

Forty-four slices are stacked into one volume.

STEP 3

Image Plus Details



2 files · gui_niftis/

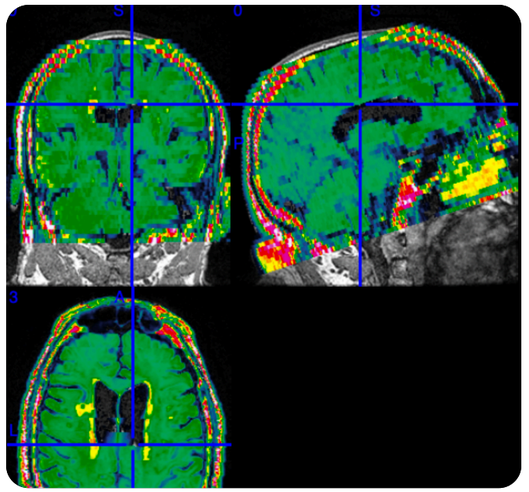
CONCEPT · NIFTI + JSON SIDECAR

```
sub-001_FLAIR_GUI.json          sub-001_MPRAGE_GUI.json
sub-001_FLAIR_GUI.nii.gz        sub-001_MPRAGE_GUI.nii.gz
```

The image travels with a record of how it was taken.

STEP 4

Look Before You Analyze



check_conversion.sh

CONCEPT · QUALITY CHECK

```
Checking Scan Converter Game outputs...

Found MPRAGE NIfTI output:
script_niftis/sub-001_MPRAGE.nii.gz

Found MPRAGE JSON output:
```

Open it and run the checker before analysis.

FROM THE SCANNER

READY FOR ANALYSIS